Materials Science Optimization Benchmark Dataset for High-dimensional, Multi-objective, Multi-fidelity Optimization of CrabNet Hyperparameters

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Abstract

Benchmarks are crucial for driving progress in scientific disciplines. To be effective, benchmarks should closely mimic real-world tasks while being computationally efficient, allowing for accessibility and repeatability. Developing surrogate models that can be indistinguishable from the ground-truth observation within the explored dataset bounds dramatically reduces the computational burden of running benchmarks without sacrificing quality, but this requires a large amount of initial data. In the fields of materials science and chemistry, relevant optimization tasks can be challenging due to their complexity, which includes hierarchical, noisy, multi-fidelity, multi-objective, high-dimensional, and non-linearly correlated variables. Additionally, they may include mixed numerical and categorical variables that are subject to linear and non-linear constraints. Simulating or experimentally verifying such tasks can be difficult, which is why benchmarks are essential. This study aimed to overcome these challenges by generating 173,219 quasi-random hyperparameter combinations across 23 hyperparameters and using them to train CrabNet on the Matbench experimental band gap dataset (computational runtime: 387 RTX-2080-Ti GPU days). The results were stored in a free-tier shared MongoDB Atlas dataset, creating a regression dataset that maps hyperparameter combinations to metrics such as MAE, RMSE, computational runtime, and model size. Heteroskedastic noise was incorporated into the regression dataset, and bad hyperparameter combinations were excluded; percentile ranks were computed within each group of identical parameter sets to capture heteroskedastic noise rather than assuming Gaussian noise. This approach can be applied to other benchmark datasets, bridging the gap between optimization benchmarks with low computational overhead and realistically complex, real-world optimization scenarios.

Keywords: Bayesian optimization, benchmark dataset, CrabNet, high-dimensional optimization, hyperparameter optimization, multi-fidelity

1 Introduction

Industry-relevant optimization tasks in the physical sciences are often "hierarchical, noisy, multi-fidelity [1–4], multi-objective [5–7], high-dimensional [8–11], and non-linearly correlated while exhibiting mixed numerical and categorical variables subject to linear [12] and non-linear constraints." [12–14] Existing benchmark datasets and platforms [15–22], while very useful, typically are single-objective, single-fidelity, low-dimensional, and ignore or simplify the influence of noise. The inclusion of heteroskedastic noise in a surrogate model enables us to establish a "Turing test" scenario where the surrogate model is virtually identical to the actual simulation. This approach bridges the gap between low-cost surrogate function evaluations using benchmark datasets and costly, real-world objective function evaluations by considering a multi-objective, multi-fidelity, and high-dimensional task while accounting for heteroskedastic noise.

The broader context for this work is the rapid rise of self-driving laboratories (SDLs) for autonomous chemical and materials discovery [23, 24], where realistic benchmark surrogates are needed to develop, compare, and de-risk the adaptive-design algorithms that drive such platforms. CrabNet itself has continued to evolve, with recent attention-based composition models extending the original architecture with structural information from crystal graphs [25]; making the hyperparameter landscape of the original CrabNet model available as a cheap surrogate enables direct, repeatable comparison of optimization strategies that might otherwise be evaluated only on the latest model.

2 Value of the data

- The dataset is valuable for benchmarking adaptive design methods applied to high-dimensional, constrained, multi-fidelity optimization tasks.
- Practitioners of optimization in the physical sciences can leverage this dataset to simulate real-world materials optimization tasks, such as alloy discovery, and achieve improved performance.

- The dataset is also useful in gaining insights into the hyperparameter optimization strategies used for developing compositionally restricted material property prediction models.

3 Data description

The regression dataset contains hyperparameter sets (including repeats) spanning twenty-three hyperparameters and their corresponding MAE, RMSE, computational runtimes, and model size for training CrabNet.

For histogram data for the number of successful repeats, see Figure 1. For histograms of the mean absolute error, root-mean-square error, runtime, and model size, see Figures 2, 3, 4, and 5, respectively.

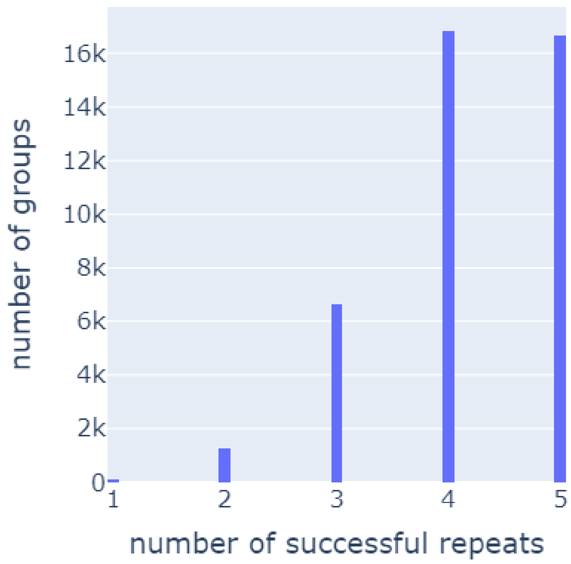

Fig. 1 Histogram of number of parameter groups vs. number of successful repeats within a given group. The lowest number of repeats for a parameter set is 1, with approximately 2.6 repeats on average.

4 Specifications table

5 Experimental design, materials and methods

A vast number of CrabNet models, totaling 173,219, were trained with different hyperparameter combinations using the Ax platform's quasi-random Sobol sampling function to generate unique parameter combinations. While there can be other uses, this dataset is primarily intended as a multi-objective, multi-fidelity, high-dimensional

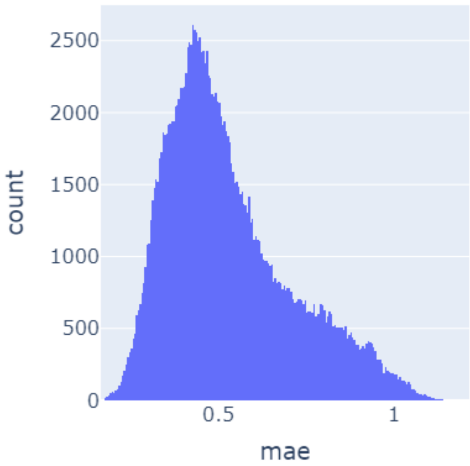

Fig. 2 Histogram of number of training runs vs. mean absolute error using CrabNet on the Matbench experimental band gap task.

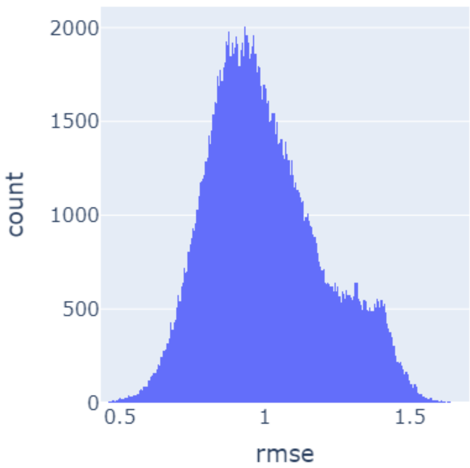

Fig. 3 Histogram of number of training runs vs. root-mean-square error using CrabNet on the Matbench experimental band gap task.

benchmark dataset for formulation-based optimization scenarios by scaling each of the numerical parameters to the range of 0 to 1 and applying a contrived constraint that the sum of all parameters must equal one. To realistically capture noise in the benchmark dataset, simulations were repeated for each quasi-random parameter combination. To improve efficiency and reduce latency, hyperparameter sets (including repeats) were shuffled and divided into batches, then sent to a high-performance computing environment for asynchronous evaluation. Despite some results not being

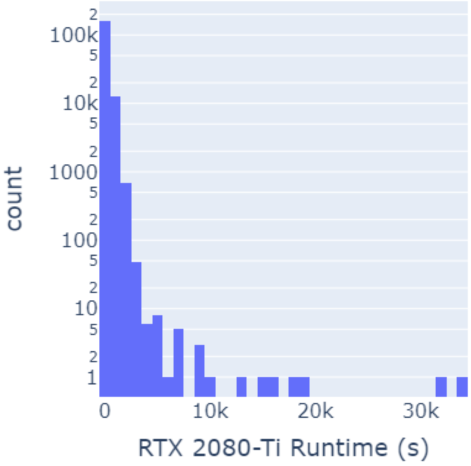

Fig. 4 Histogram of number of training runs vs. GPU runtime on an RTX 2080-Ti using CrabNet on the Matbench experimental band gap task. The y-axis is log-scaled.

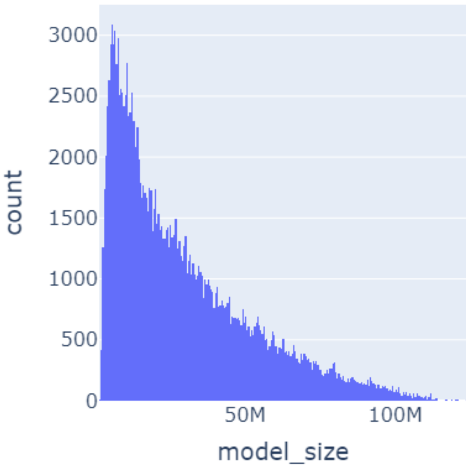

Fig. 5 Histogram of number of training runs vs. model size using CrabNet on the Matbench experimental band gap task.

completed due to either timeout or preemption, this trade-off was deemed reasonable for the efficiency and completion gains.

Results were logged to a free-tier MongoDB Atlas database and then aggregated and prepared as machine-learning-ready datasets via Python in Jupyter notebooks. For implementation details, see https://github.com/sparks-baird/matsci-opt-benchmarks/tree/main/scripts/crabnet_hyperparameter and https://github.com/sparks-baird/matsci-opt-benchmarks/tree/main/notebooks/crabnet_hyperparameter. The trained surrogate has been deployed as an interactive

Table 1 Dataset specifications.

Subject	Computational materials science
Specific subject area	Composition-based experimental band gap prediction
Type of data	Table; Figure
How the data were acquired	Executed CrabNet v2.0.8 (https://github.com/sparks-baird/CrabNet) for each of the five folds of the Matbench experimental band gap task (https://matbench.materialsproject.org/). Python orchestration in `scripts/crabnet_hyperparameter/crabnet_hyperparameter_submitit.py` was used to dispatch jobs to SLURM via Submitit on the University of Utah's Center for High-performance Computing (CHPC); results were logged in JSON via the MongoDB Data API. Source release at v0.2.1 (https://doi.org/10.5281/zenodo.7694289).
Data format	Analyzed; Filtered; Raw
Description of data collection	The Ax Platform was used to perform a quasi-random Sobol sampling of 65,536 parameter combinations, varying 23 hyperparameters within a constrained search space, with 5 repeats resulting in a total of 327,680 training runs. Of these, 173,219 completed successfully, consuming 387 RTX-2080-Ti GPU days (˜4,614 CUDA core years), resulting in 41,550 unique sets. To rank repeat simulations, the "dense" method with `pct=True` was used in `pandas.core.groupby.GroupBy.rank`.
Data source location	Free-tier shared cluster MongoDB Atlas database.
Data accessibility	Repository: Zenodo. Identifier: 7694268. Direct URL: https://doi.org/10.5281/zenodo.7694268.

HuggingFace Space[1] and used downstream for benchmarking adaptive-design libraries such as BayBE [26] and Ax against one another[2], as well as in an Acceleration Consortium Kaggle benchmarking competition[3].

Ethics statements

There are no statements to declare.

CRediT author statement

Sterling G. Baird: Project administration, Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization. **Jeet N. Parikh**: Methodology, Software, Formal Analysis, Data Curation, Writing – Original Draft, Writing – Review & Editing. **Taylor D. Sparks**: Supervision, Funding acquisition.

[1] https://huggingface.co/spaces/AccelerationConsortium/crabnet-hyperparameter
[2] https://github.com/AccelerationConsortium/baybe-multi-task-bo
[3] https://www.kaggle.com/datasets/acceleration-consortium/crabnet-optimization-challenge-dataset

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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